

EX. No. 19 – High-Level Programming Extensions – Triggers

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MySQL 8.0 Command Line Cli x + v

mysql> CREATE DATABASE IF NOT EXISTS TriggerDB;
Query OK, 1 row affected (0.04 sec)

mysql> USE TriggerDB;
Database changed
mysql> DROP TABLE IF EXISTS Employees;
Query OK, 0 rows affected, 1 warning (0.04 sec)

mysql>
mysql> CREATE TABLE Employees (
->   EmpID INT AUTO_INCREMENT PRIMARY KEY,
->   EmpName VARCHAR(50),
->   Salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER MinSalaryBeforeInsert
-> BEFORE INSERT ON Employees
-> FOR EACH ROW
-> BEGIN
->   IF NEW.Salary < 30000 THEN
->     SET NEW.Salary = 30000;
->   END IF;
-> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employees (EmpName, Salary) VALUES ('Alice', 25000);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employees (EmpName, Salary) VALUES ('Bob', 40000);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Employees;
+-----+-----+-----+
| EmpID | EmpName | Salary |
+-----+-----+-----+
```

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mysql> CREATE TABLE SalaryLog (
->   LogID INT AUTO_INCREMENT PRIMARY KEY,
->   EmpID INT,
->   OldSalary DECIMAL(10,2),
->   NewSalary DECIMAL(10,2),
->   ChangeDate DATETIME
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER SalaryUpdateLog
-> AFTER UPDATE ON Employees
-> FOR EACH ROW
-> BEGIN
->   IF OLD.Salary <> NEW.Salary THEN
->     INSERT INTO SalaryLog(EmpID, OldSalary, NewSalary, ChangeDate)
->     VALUES (OLD.EmpID, OLD.Salary, NEW.Salary, NOW());
->   END IF;
-> END$$
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER ;
mysql> UPDATE Employees SET Salary = 35000 WHERE EmpName = 'Alice';
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Employees SET Salary = 45000 WHERE EmpName = 'Bob';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM SalaryLog;
+-----+-----+-----+-----+-----+
| LogID | EmpID | OldSalary | NewSalary | ChangeDate |
+-----+-----+-----+-----+-----+
| 1 | 1 | 30000.00 | 35000.00 | 2025-10-08 21:33:49 |
| 2 | 2 | 40000.00 | 45000.00 | 2025-10-08 21:33:49 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```